



The Treasures of Gaberoun

Ahmed is an explorer leading an expedition to the **Gaberoun Oasis** in the heart of the Sahara. Recent surveys suggest that beneath the dunes lies a hidden network of **Special Soil**—a rare, mineral-rich earth used for ancient pottery.

The exploration area is represented as an $n \times m$ grid. The first row is at the surface level, and the n -th row is the deepest point. Each cell in the grid is either:

- `.` : **Special Soil** (valuable)
- `#` : **Normal Soil** (not valuable)

To extract the Special Soil, Ahmed has k **vertical pipes**, each of length 1. A cell containing Special Soil is considered **reachable** (and thus collectable) if:

1. A pipe passes through it.
2. There is a set of Special Soil cells such that the distance between each one is equal to 1 (they are connected horizontally or vertically), and there is a pipe passes through at one of them.

Let's call the set of Special Soil cells mentioned in the last point **Component**.

Pipes can overlap in the same column, but each set of pipes can only have at most one component of Special Soil cells

Ahmed wants to maximize his find. What is the maximum number of Special Soil cells he can collect using at most k vertical pipes?

Constraints

- $n \times m \leq 5 \cdot 10^5$
- $k \leq n \times m$

I/O

Input

- The first line contains three integers n, m, k representing the size of the grid and the number of pipes available.
- The next n lines each contain a string of length m , consisting of the characters `.` and `#`.

Output

- Print a single integer: the maximum number of Special Soil cells Ahmed can reach using at most k pipes.

Subtasks

Subtask	Constraints	Points
1	$m = 1, n \leq 1000$	5
2	$n = 1, m \leq 1000$	6
3	$n = 1, m \leq 5 \cdot 10^5$	14
4	$1 \leq n, m \leq 6$	7
5	$n \times m \leq 1000$	23
6	$m, k \leq 1000$	19
7	No additional constraints	26

Example

Input:

```
3 3 1
.#.
...
#.#
```

Output:

```
6
```

Explanation: The Special Soil cells form one large connected component of size 6. Since $k = 1$, Ahmed can place a pipe in any cell belonging to the large component to collect all 6 cells.

Input:

```
3 1 3
.
#
.
```

Output:

1

Explanation: To reach the topmost Special Soil cell, you need to use only 1 pipe, to use the bottommost Special Soil cell, you need to use 3 pipes, you can't have both sense $1 + 3 = 4 > 3$

Input:

```
5 5 5
##.##
...##
.####
##.##
##...
```

Output:

9

Explanation: To get the first component, use only one pipe and you will have 5 Special Soil cells, then you need 4 pipes to reach the second component, resulting in having all the Special Soil cells.